Def. Given two tangent vectors V_p and W_p at $p \in \mathbb{R}^3$, define

$$V_p \times W_p \overset{\text{def}}{=} \begin{vmatrix} 1 & 0 & 0 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{vmatrix} \cdot (e_1)_p + \begin{vmatrix} 0 & 1 & 0 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{vmatrix} \cdot (e_2)_p + \begin{vmatrix} 0 & 0 & 1 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{vmatrix} \cdot (e_3)_p .$$

It is called the Cross Product of V_p and W_p.

Prop. $V_p \times W_p = 0$ if and only if V_p and W_p are linearly dependent.

Proof. Suppose that $V_p \times W_p = 0$. Then, for any $(u_1, u_2, u_3) \in \mathbb{R}^3$,

$$\det \begin{bmatrix} u_1 & u_2 & u_3 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{bmatrix} = 0.$$ So, taking $(u_1, u_2, u_3) \in \mathbb{R}^3 \setminus \text{Span}\{V, W\}$, we obtain the result. (If V and W are linearly independent, then the determinant cannot be 0.)

Conversely, if V_p and W_p are linearly dependent, then regardless of the first row, $\det = 0$. ∎

Lemma. Given V_p and W_p in $T_p(\mathbb{R}^3)$, $V_p \times W_p$ is orthogonal to both V_p and W_p, and

$$\|V_p \times W_p\|^2 = (V_p \cdot V_p)\cdot(W_p \cdot W_p) - (V_p \cdot W_p)^2 .$$

In other say, $\|V \times W\| = \|V\| \cdot \|W\| \cdot \sin\theta$

Proof. By the linearity of the determinant,

$$V \cdot (V \times W) = \begin{vmatrix} V_1 & V_2 & V_3 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{vmatrix} = 0 \quad \text{and} \quad W \cdot (V \times W) = \begin{vmatrix} W_1 & W_2 & W_3 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{vmatrix} = 0.$$

Therefore, the orthogonality is proved. And, to show the second assertion,

$$(V \cdot V)\cdot(W \cdot W) - (V \cdot W)^2 = \left(V_1^2 + V_2^2 + V_3^2\right)\left(W_1^2 + W_2^2 + W_3^2\right) - \left(V_1 W_1 + V_2 W_2 + V_3 W_3\right)^2$$

$$= \sum_{1 \le i, j \le 3} V_i^2 \cdot W_j^2 - \left(\sum_{1 \le i, j \le 3} V_i W_i V_j W_j \right)$$

$$= \sum_{1 \le i, j \le 3} V_i^2 \cdot W_j^2 - \left(\sum_{i=1}^{3} V_i^2 W_i^2 + 2 \cdot \sum_{1 \le i < j \le 3} V_i W_i V_j W_j \right)$$

$$= \sum_{1 \le i < j \le 3} V_i^2 \cdot W_j^2 - 2 \cdot \sum_{1 \le i < j \le 3} V_i W_i V_j W_j$$

Meanwhile, $\|V \times W\|^2 = (V_2 W_3 - V_3 W_2)^2 + (V_1 W_3 - V_3 W_1)^2 + (V_1 W_2 - V_2 W_1)^2$

$$= V_2^2 W_3^2 + V_3^2 W_2^2 - 2 V_2 W_2 V_3 W_3$$
$$+ V_1^2 W_3^2 + V_3^2 W_1^2 - 2 V_1 W_1 V_3 W_3$$
$$+ V_1^2 W_2^2 + V_2^2 W_1^2 - 2 V_1 W_1 V_2 W_2$$

These are same. ∎